

Tandberg Terminal Keyboard Documentation

Overview

This document describes the keyboard layouts for two Tandberg terminal models:

- **TDV-2200/9S** (ND320 Norsk Data) - Layout 531300
- **TDV-2215** (General Purpose ASCII) - Layout 129903

Keyboard Layout Structure

Each keyboard uses a multi-layer scan code system with the following structure:

- **0x000-0x07F**: Key flags (control bits)
- **0x080-0x0FF**: Normal key values
- **0x100-0x17F**: Shift held key values
- **0x180-0x1FF**: Ctrl held key values

Flag System

The Flags byte controls how each key is processed:

```
Flags: xxxxxxxx
| | | | | | | |
| | | | | | | |
| | | | | | | | Bit 0: Use key with Normal (1=enabled, 0=disabled)
| | | | | | | | Bit 1: Reserved
| | | | | | | | Bit 2: Reserved
| | | | | | | | Bit 3: Reserved
| | | | | | | | Bit 4: Prefix CTRL-key with 0xFF (1=yes, 0=no)
| | | | | | | | Bit 5: Treat Caps as Normal (1=yes, 0=no)
| | | | | | | | Bit 6: Reserved
| | | | | | | | Bit 7: Disable key-repeat (1=disabled, 0=enabled)
```

Common flag values:

- **0x0F** (00001111): Standard alphanumeric keys
- **0x1F** (00011111): Special keys requiring FF prefix for Ctrl
- **0x37** (00110111): Function keys with special behavior

Complete Key Mappings

TDV-2215 Layout (129903) - Recommended for Implementation

Position	Key Name	Normal	Shift	Ctrl	Flags	SDL Mapping
Letters A-Z						
C1	A	0x61	0x41	0x01	0x0F	SDLK_a
B5	B	0x62	0x42	0x02	0x0F	SDLK_b
B3	C	0x63	0x43	0x03	0x0F	SDLK_c
C3	D	0x64	0x44	0x04	0x0F	SDLK_d
D3	E	0x65	0x45	0x05	0x0F	SDLK_e
C4	F	0x66	0x46	0x06	0x0F	SDLK_f
C5	G	0x67	0x47	0x07	0x0F	SDLK_g
C6	H	0x68	0x48	0x08	0x0F	SDLK_h
D8	I	0x69	0x49	0x09	0x0F	SDLK_i
C7	J	0x6A	0x4A	0x0A	0x0F	SDLK_j
C8	K	0x6B	0x4B	0x0B	0x0F	SDLK_k
C9	L	0x6C	0x4C	0x0C	0x0F	SDLK_l
B7	M	0x6D	0x4D	0x0D	0x0F	SDLK_m
B6	N	0x6E	0x4E	0x0E	0x0F	SDLK_n
D9	O	0x6F	0x4F	0x0F	0x0F	SDLK_o
D10	P	0x70	0x50	0x10	0x0F	SDLK_p
D1	Q	0x71	0x51	0x11	0x0F	SDLK_q
D4	R	0x72	0x52	0x12	0x0F	SDLK_r
C2	S	0x73	0x53	0x13	0x0F	SDLK_s
D5	T	0x74	0x54	0x14	0x0F	SDLK_t
D7	U	0x75	0x55	0x15	0x0F	SDLK_u
B4	V	0x76	0x56	0x16	0x0F	SDLK_v
D2	W	0x77	0x57	0x17	0x0F	SDLK_w
B2	X	0x78	0x58	0x18	0x0F	SDLK_x
D6	Y	0x79	0x59	0x19	0x0F	SDLK_y
B1	Z	0x7A	0x5A	0x1A	0x0F	SDLK_z
Numbers 0-9						
E10	0 =	0x30	0x3D	0x00	0x17	SDLK_0
E1	1 !	0x31	0x21	0x00	0x17	SDLK_1
E2	2 "	0x32	0x22	0x00	0x17	SDLK_2
E3	3 #	0x33	0x23	0x00	0x17	SDLK_3
E4	4 \$	0x34	0x24	0x00	0x17	SDLK_4
E5	5 %	0x35	0x25	0x00	0x17	SDLK_5
E6	6 &	0x36	0x26	0x00	0x17	SDLK_6

Position	Key Name	Normal	Shift	Ctrl	Flags	SDL Mapping
E7	7 /	0x37	0x2F	0x00	0x17	SDLK_7
E8	8 (	0x38	0x28	0x00	0x17	SDLK_8
E9	9)	0x39	0x29	0x00	0x17	SDLK_9
Navigation & Control						
C48	Up	0xB1	—	0x00	0x17	SDLK_UP
A48	Down	0xB2	—	0x00	0x17	SDLK_DOWN
B49	Right	0xB3	—	0x00	0x17	SDLK_RIGHT
B47	Left	0xB4	—	0x00	0x17	SDLK_LEFT
B48	Home	0xB8	—	0x00	0x37	SDLK_HOME
Special Keys						
A5	Space	0x20	—	0x00	0x1F	SDLK_SPACE
E13	Backspace	0x08	—	0x00	0x17	SDLK_BACKSPACE
A49	Tab Right	0x09	0x99	—	0x3F	SDLK_TAB
C13	CR	0xCD	—	—	0x37	SDLK_RETURN
F53	ESC	0xDB	—	—	0x37	SDLK_ESCAPE
D13	LF	0x9A	—	—	0x17	SDLK_RETURN2
E14	DEL	0x7F	—	—	0x37	SDLK_DELETE
Function Keys P1-P8						
G1	P1	0xA0	0xA8	0x81	0x3F	SDLK_F1
G2	P2	0xA1	0xA9	0x82	0x3F	SDLK_F2
G3	P3	0xA2	0xAA	0x83	0x3F	SDLK_F3
G4	P4	0xA3	0xAB	0x84	0x3F	SDLK_F4
G5	P5	0xA4	0xAC	0x85	0x3F	SDLK_F5
G6	P6	0xA5	0xAD	0x86	0x3F	SDLK_F6
G7	P7	0xA6	0xAE	0x87	0x3F	SDLK_F7
G8	P8	0xA7	0xAF	0x88	0x3F	SDLK_F8
Numeric Keypad						
A51	0 (np)	0xE0	—	—	0x17	SDLK_KP_0
B51	1 (np)	0xE1	—	—	0x17	SDLK_KP_1
B52	2 (np)	0xE2	—	—	0x17	SDLK_KP_2
B53	3 (np)	0xE3	—	—	0x17	SDLK_KP_3
C51	4 (np)	0xE4	—	—	0x17	SDLK_KP_4
C52	5 (np)	0xE5	—	—	0x17	SDLK_KP_5
C53	6 (np)	0xE6	—	—	0x17	SDLK_KP_6
D51	7 (np)	0xE7	—	—	0x17	SDLK_KP_7

Position	Key Name	Normal	Shift	Ctrl	Flags	SDL Mapping
D52	8 (np)	0xE8	—	—	0x17	SDLK_KP_8
D53	9 (np)	0xE9	—	—	0x17	SDLK_KP_9
C54	- (np)	0xEA	—	—	0x17	SDLK_KP_MINUS
A53	. (np)	0xEB	—	—	0x17	SDLK_KP_PERIOD
B54	ENTER (np)	0xED	—	—	0x17	SDLK_KP_ENTER
System Keys						
G53	MODE	0x00	0xF8	0x00	0x34	SDLK_INSERT
G54	BREAK	0xFE	0xF9	0xF4	0x3D	SDLK_PAUSE
G52	PRINT	0x00	0xFC	0xFD	0x3C	SDLK_PRINTSCREEN

Special Key Combinations

Configuration Menu Navigation

- **Enter Config:** MODE key (0xF8)
- **Cursor Up:** Up arrow (0xB1)
- **Cursor Down:** Down arrow (0xB2)
- **Cursor Left:** Left arrow (0xB4)
- **Cursor Right:** Right arrow (0xB3)
- **Enter/Select:** Numeric keypad ENTER (0xED)
- **Exit:** ESC key (0xDB)

Key Transmission Rules

1. **Normal Keys:** Transmit the Normal value directly
2. **Shift Keys:** Transmit the Shift value directly
3. **Ctrl Keys:**
 - If Flags bit 4 is set: Send 0xFF prefix + Ctrl value
 - Otherwise: Send Ctrl value directly
4. **Special Flags:** Check bit 0 to determine if key is enabled for Normal mode

Layout Differences

TDV-2200/9S (531300) vs TDV-2215 (129903)

The main differences:

- **2200/9S:** Norwegian layout with Æ, Ø, Å characters
- **2215:** ASCII layout with standard punctuation

- **Function Keys:** Different P-key mappings
- **Special Keys:** Different system key assignments

Implementation Notes

- Use TDV-2215 layout as base for broad compatibility
- Handle modifier keys (Shift, Ctrl) properly
- Implement FF prefix for special Ctrl combinations
- Support both regular and numeric keypad ENTER keys
- Map SDL scancodes to physical key positions

Based on official Tandberg keyboard layout specifications 531300 and 129903